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Abstract

Bank customer churn, the phenomenon where customers discontinue their relationship with a financial institution, poses a significant challenge to the banking industry. This report presents a supervised machine learning approach to predict customer churn using the Bank Customer Churn Prediction dataset, which contains 10,000 records and 12 features sourced from Kaggle. The dataset exhibits a class imbalance of approximately 80:20 (non-churn to churn), which was addressed using the Synthetic Minority Oversampling Technique (SMOTE). Two classification algorithms were employed: Logistic Regression and Decision Tree Classifier. Preprocessing steps included label encoding of categorical variables, Standard Scaler normalization, and stratified train-test splitting. Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices with 10-fold stratified cross-validation. The results demonstrate that [Best Model] achieved superior performance with an accuracy of [X]% and an F1-score of [X], outperforming [Other Model]. Feature importance analysis revealed that age, balance, and number of products were the most influential predictors of churn. The findings suggest that machine learning can be effectively applied to support proactive customer retention strategies in banking.

1. Introduction

1.1 Background

The banking industry operates in an increasingly competitive environment where customer retention is critical to profitability. Customer churn, defined as the voluntary exit of a customer from a bank's services, has direct implications for revenue and market share. Research indicates that acquiring a new customer costs significantly more than retaining an existing one, making churn prediction a high-priority business intelligence task.
Machine learning has emerged as a powerful tool for churn prediction, enabling banks to proactively identify at-risk customers and deploy targeted retention interventions. Supervised classification algorithms, in particular, have demonstrated strong performance in modelling the complex, non-linear relationships between customer attributes and churn behaviour.

1.2 Problem Statement

Despite the availability of rich customer data, many banks rely on reactive rather than predictive approaches to churn management. The core problem addressed in this report is given a set of demographic and behavioural features, can a supervised machine learning model accurately classify whether a bank customer is likely to churn? The dataset used presents an additional challenge of class imbalance, where churned customers represent only approximately 20% of the total population, which can bias model learning toward the majority class.

1.3 Objectives

• To explore and understand the Bank Customer Churn dataset through exploratory data analysis.
•To apply appropriate preprocessing techniques including encoding, scaling, and SMOTE to prepare the data for modelling.

- To train and evaluate two supervised learning classifiers, Logistic Regression and Decision Tree, on the preprocessed dataset.
- To compare model performance using standard classification metrics and ROC-AUC curves.
- To identify the most influential features driving customer churn.

2. Literature Review

2.1 Dataset Usage and Research Trends

The Bank Customer Churn dataset is widely used in machine learning research for binary classification tasks, particularly in the banking and financial services sector. It contains structured customer data with a mixture of numerical, categorical, and binary features, along with a moderately imbalanced target variable. Due to these characteristics, it serves as a practical benchmark for evaluating classification algorithms and preprocessing techniques.

Recent research trends in churn prediction have shifted from traditional statistical models toward more advanced machine learning approaches, including ensemble methods and deep learning. However, for structured tabular datasets, classical models such as Logistic Regression and Decision Trees remain highly relevant due to their interpretability and competitive performance.

2.2 Related Studies

Tran et al. (2023) investigated customer churn prediction in the banking sector using multiple machine learning classification models. Their findings showed that machine learning techniques can effectively identify at-risk customers and improve retention strategies, with tree-based models demonstrating strong predictive performance.

Peng et al. (2023) focused on churn prediction and model interpretability, highlighting the importance of understanding feature contributions in classification models. The study emphasized that features such as customer age, account balance, and product usage play a significant role in predicting churn behavior.

Ahmad et al. (2024) conducted a comparative analysis of machine learning models for churn prediction and found that nonlinear models generally outperform linear models in capturing complex customer behavior patterns. Their results support the use of Decision Tree models for improved classification performance.

Berrada Chakour et al. (2026) explored the use of SMOTE-based resampling techniques on imbalanced churn datasets. Their study demonstrated that SMOTE significantly improves minority class prediction without sacrificing overall model stability, making it an effective technique for handling class imbalance.

Fernández et al. (2018) provided a comprehensive review of SMOTE and its applications in imbalanced datasets. The study confirmed that oversampling techniques are essential for improving recall and F1-score in classification problems where the minority class is of primary interest.

2.3 Summary of Literature

Study	Year	Key Findings
Tran et al.	2023	Machine learning models effectively predict churn
Peng et al.	2023	Age, balance, and product usage are key features
Ahmad et al.	2024	Nonlinear models outperform linear models
Berrada Chakour et al.	2026	SMOTE improves minority class prediction
Fernández et al.	2018	Oversampling improves F1-score and recall

2.4 Justification of Approach

Based on the literature, Logistic Regression is selected as a baseline model due to its simplicity and interpretability, while Decision Tree is chosen for its ability to capture nonlinear relationships in customer behavior. Additionally, the use of SMOTE for handling class imbalance is supported by multiple studies, ensuring improved performance in predicting the minority (churn) class.

2. Dataset Description

2.1 Dataset Source

The dataset used in this project is the Bank Customer Churn Prediction dataset, publicly available on Kaggle at the following URL:

https://www.kaggle.com/datasets/shantanudhakadd/bank-customer-churn-prediction

2.2 Dataset Overview

The dataset contains 10,000 customer records collected from a European bank, with 12 attributes per record including demographic information, account details, and a binary churn indicator. The target variable is 'churn', where a value of 1 indicates the customer has left the bank and 0 indicates they remain. The dataset exhibits a notable class imbalance, with approximately 79.6% non-churned and 20.4% churned customers.

2.3 Feature Description Table

Feature Name	Data Type	Description
customer_id	Integer	Unique identifier for each customer (dropped in preprocessing)
credit_score	Integer	Customer's credit score
country	Categorical	Customer's country of residence (France, Germany, Spain)
gender	Categorical	Customer's gender (Male, Female)
age	Integer	Customer's age in years
tenure	Integer	Number of years the customer has been with the bank
balance	Float	Customer's account balance
products_number	Integer	Number of bank products held by the customer
credit_card	Binary	Whether the customer has a credit card (1 = Yes, 0 = No)

active_member	Binary	Whether the customer is an active member (1 = Yes, 0 = No)
estimated_salary	Float	Customer's estimated annual salary
churn	Binary	Target variable whether customer churned (1 = Yes, 0 = No)

Table 1: Feature descriptions for the Bank Customer Churn Prediction dataset

3. Exploratory Data Analysis (EDA)

3.1 Data Understanding

Initial exploration of the dataset confirmed 10,000 rows and 12 columns with no missing values across any feature. The dataset contains a mix of numerical features (credit_score, age, tenure, balance, estimated_salary), binary features (credit_card, active_member, churn), and categorical features (country, gender, products_number). The customer_id column was identified as a non-predictive identifier and was dropped prior to modelling.

3.2 Data Distribution

The class distribution analysis revealed a significant imbalance in the target variable, with 7,963 non-churned customers (79.63%) versus 2,037 churned customers (20.37%). This imbalance necessitated the application of SMOTE during preprocessing to prevent the models from being biased toward predicting the majority class.

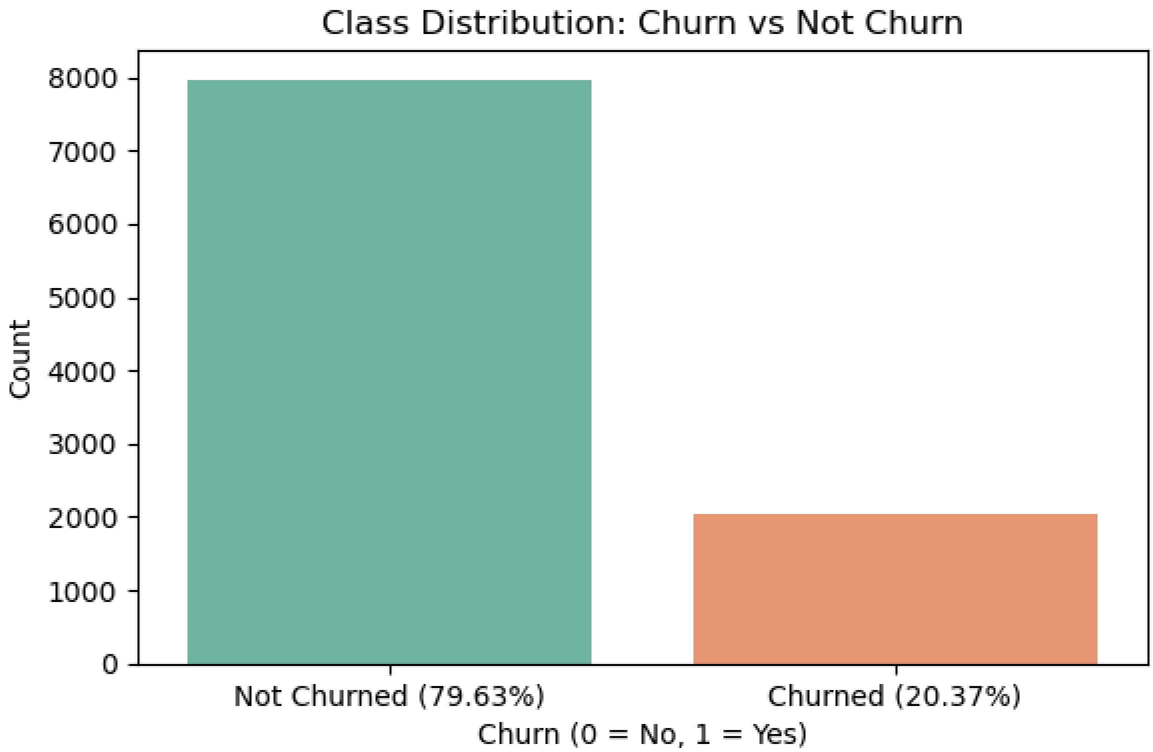


Figure 1: Class distribution of the target variable (Churn vs Not Churn)

Histograms of numerical features revealed that balance has a bimodal distribution with a large spike at zero (many customers hold zero balance), while age follows an approximately normal distribution skewed slightly right. Credit score, tenure, and estimated salary are relatively uniformly distributed.

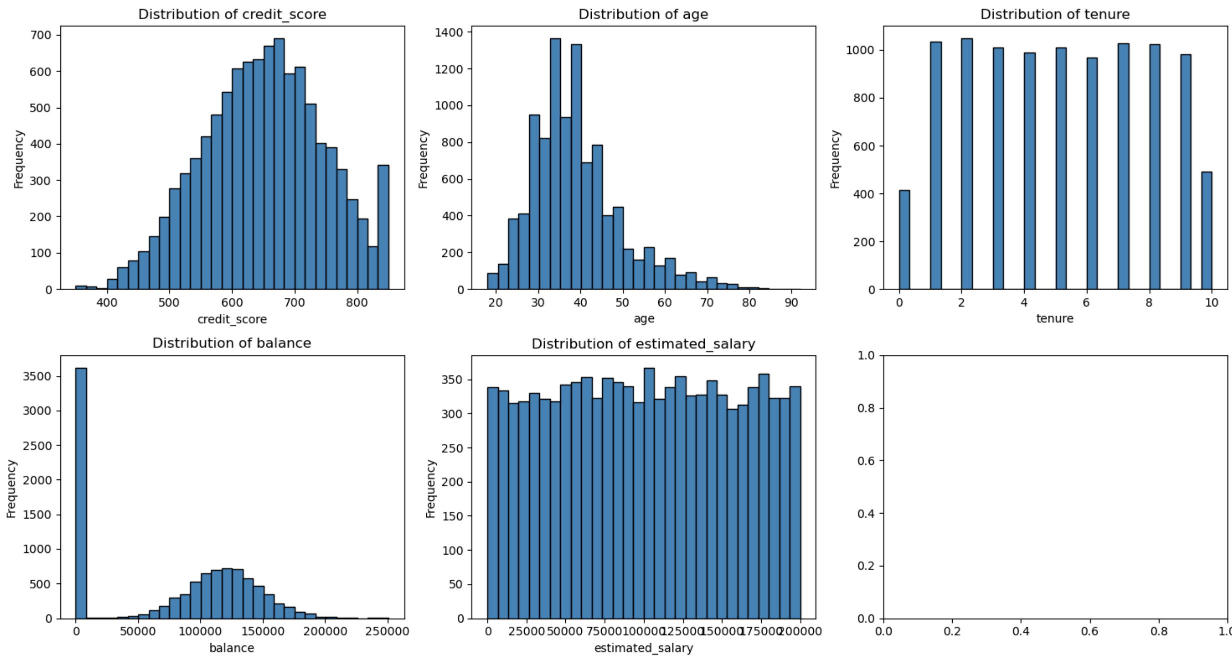


Figure 2: Distribution of numerical features

3.3 Relationships

Analysis of churn rates by country revealed that customers from Germany have a notably higher churn rate compared to France and Spain. Gender-based analysis showed a slightly higher churn tendency among female customers. The correlation heatmap indicated that age has the strongest positive correlation with churn, while being an active member shows a negative correlation, suggesting active engagement reduces churn likelihood.

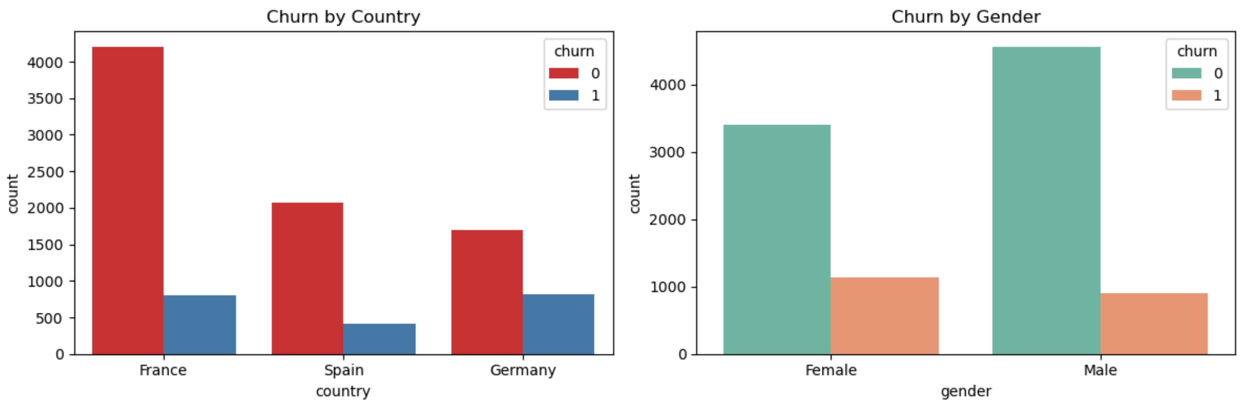
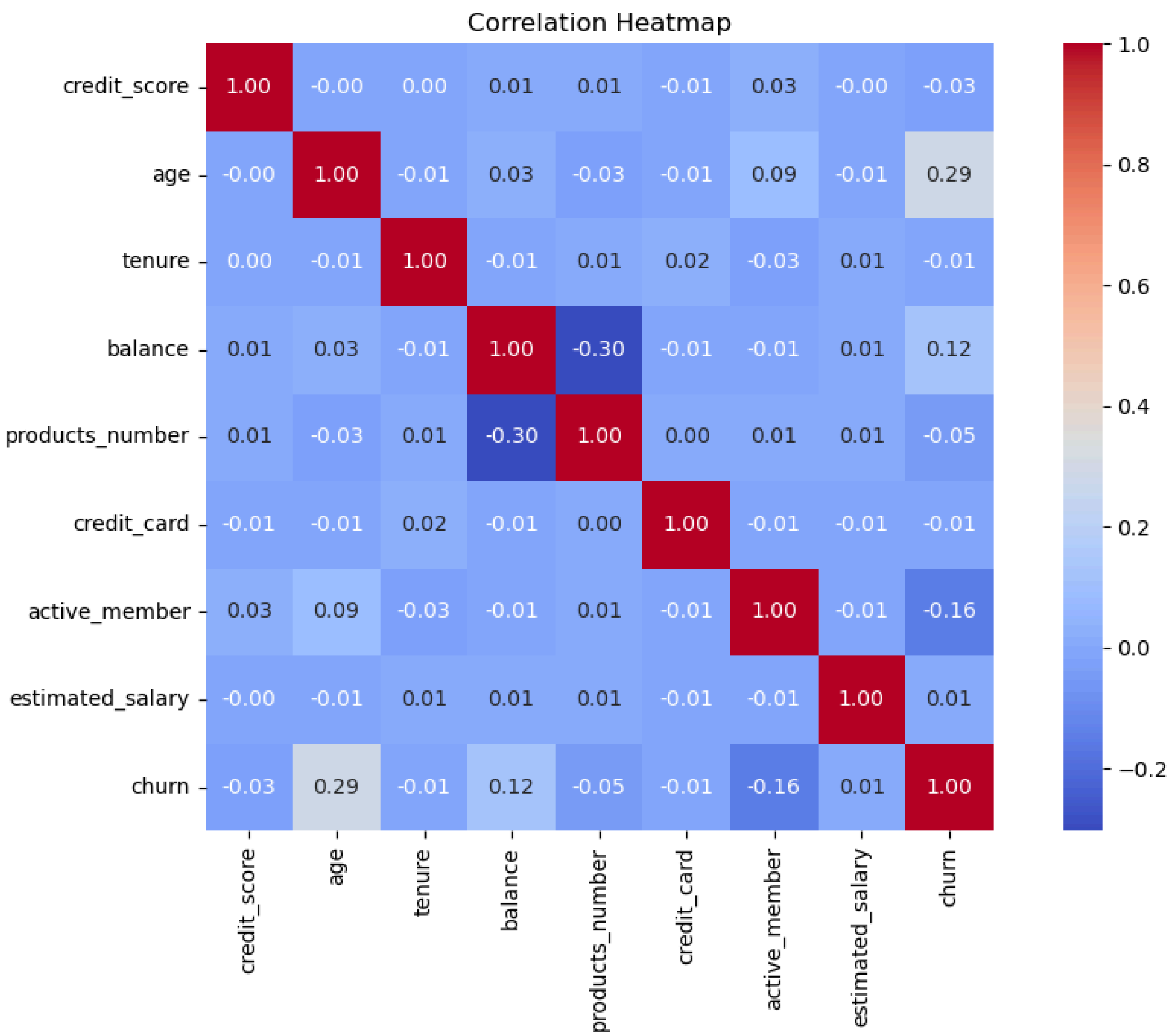


Figure 3: Churn rate by country (left) and gender (right)



	credit_score	age	tenure	balance	products_number	credit_card	active_member	estimated_salary	churn
credit_score	1.00	-0.00	0.00	0.01	0.01	-0.01	0.03	-0.00	-0.03
age	-0.00	1.00	-0.01	0.03	-0.03	-0.01	0.09	-0.01	0.29
tenure	0.00	-0.01	1.00	-0.01	0.01	0.02	-0.03	0.01	-0.01
balance	0.01	0.03	-0.01	1.00	-0.30	-0.01	-0.01	0.01	0.12
products_number	0.01	-0.03	0.01	-0.30	1.00	0.00	0.01	0.01	-0.05
credit_card	-0.01	-0.01	0.02	-0.01	0.00	1.00	-0.01	-0.01	-0.01
active_member	0.03	0.09	-0.03	-0.01	0.01	-0.01	1.00	-0.01	-0.16
estimated_salary	-0.00	-0.01	0.01	0.01	0.01	-0.01	-0.01	1.00	0.01
churn	-0.03	0.29	-0.01	0.12	-0.05	-0.01	-0.16	0.01	1.00

Figure 4: Correlation heatmap of numerical features

3.4 Key Findings

- The dataset is imbalanced, approximately 80:20 non-churn to churn ratio.
- Age is the most positively correlated feature with churn; older customers are more likely to leave.
- Active membership status is negatively correlated with churn, inactive members churn more.
- German customers exhibit disproportionately higher churn rates compared to French and Spanish customers.
- A large proportion of customers hold zero account balance, creating a bimodal distribution in the balance feature.

4. Data Preprocessing

4.1 Handling Missing Values

A thorough inspection of the dataset using isnull().sum() confirmed that there are no missing values in any column. Therefore, no imputation or row removal was required. This contributes to the data quality and reliability of subsequent analyses.

4.2 Encoding Categorical Variables

Two categorical columns required encoding prior to modelling. The 'gender' column (Male/Female) was encoded using LabelEncoder, converting Female to 0 and Male to 1. The 'country' column (France/Germany/Spain) was similarly encoded using LabelEncoder, mapping France to 0, Germany to 1, and Spain to 2. This transformation ensures all features are in a numerical format compatible with scikit-learn classifiers.

4.3 Feature Scaling

StandardScaler was applied to normalize all features after the train-test split to prevent data leakage. The scaler was fitted exclusively on the training set and subsequently used to transform the test set. This ensures that the test set statistics do not influence the scaling parameters, preserving the integrity of model evaluation. Scaling is particularly important given the large range differences between features such as 'balance' (0 to ~250,000) and 'tenure' (0 to 10).

4.4 Outlier Handling

A review of the descriptive statistics and distribution plots did not reveal extreme outliers requiring removal. The balance feature contains many zero values, which is a genuine characteristic of the dataset (customers with no account balance) rather than an error. All values were retained as they represent valid customer states.

4.5 Handling Imbalanced Data

The class imbalance (79.63% non-churn vs 20.37% churn) was addressed using the Synthetic Minority Oversampling Technique (SMOTE). SMOTE generates synthetic samples for the minority class by interpolating between existing minority class instances rather than simply duplicating them, resulting in a more diverse and robust resampled training set. SMOTE was applied exclusively to the training data after the train-test split to prevent information leakage from the test set into the resampling process. The resulting training set achieved a balanced 50:50 class distribution.

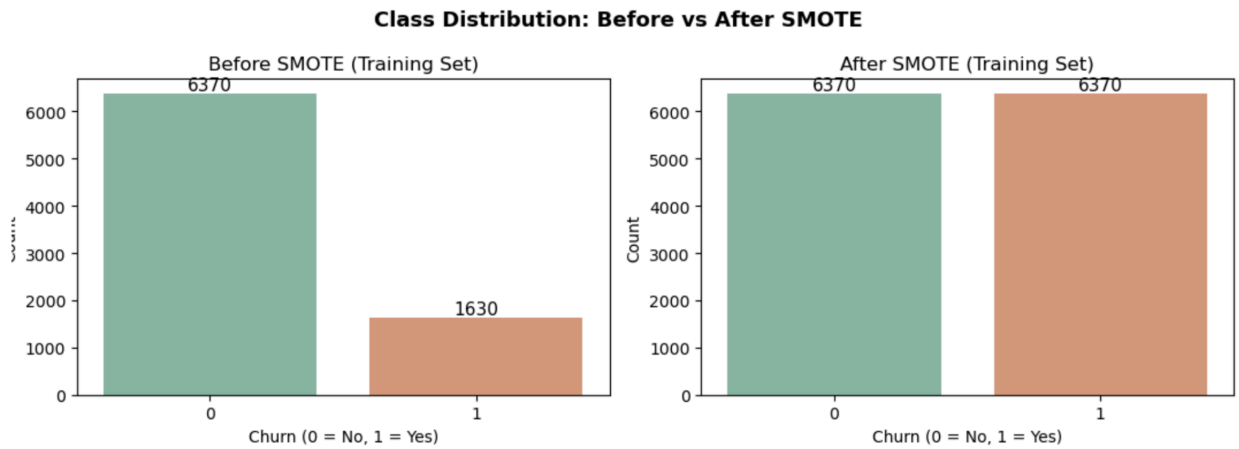


Figure 5: Class distribution before and after applying SMOTE

4.6 Final Dataset

After preprocessing, the final dataset used for modelling consists of 11 features (customer_id removed) with all categorical variables encoded and all continuous features normalized. The training set was balanced using SMOTE, while the test set retained the original class distribution to simulate real-world evaluation conditions. The data was split with an 80:20 train-test ratio using stratified sampling.

5. Dimension Reduction

5.1 Method Used

Feature selection by domain knowledge and correlation analysis was applied as the dimension reduction strategy, rather than Principal Component Analysis (PCA). The customer_id column was explicitly removed as it serves solely as a record identifier with no predictive value.

5.2 Justification

PCA was not applied in this project for two reasons. First, the dataset contains only 11 predictive features after removing customer_id, which is already a compact feature space. Second, PCA transforms features into components that are linear combinations of the originals, making it difficult to interpret feature importance, a key requirement in this analysis. Retaining the original features allows direct interpretation of the Decision Tree's feature importances, which is essential for the discussion section.

5.3 Results

All 11 remaining features were retained. The correlation heatmap confirmed there is no extreme multicollinearity between features that would necessitate further reduction. The most informative features, as later confirmed by the Decision Tree feature importance analysis, are age, balance, products_number, and active_member.

5.4 Visualization

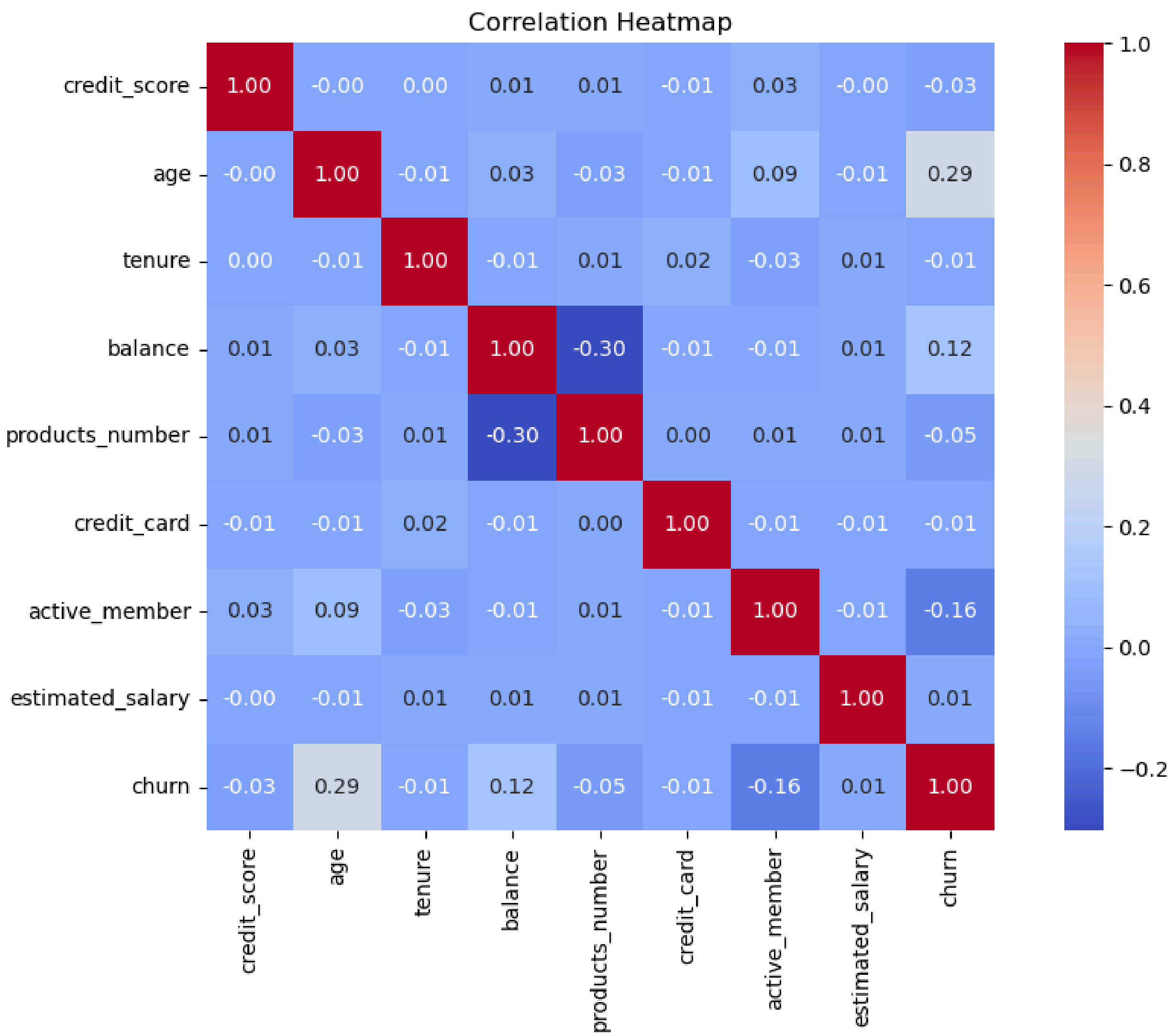


Figure 6: Correlation heatmap used to assess feature relationships and justify feature retention

6. Supervised Learning Models

6.1 Model 1 - Logistic Regression

Logistic Regression is a linear classification algorithm that models the probability of a binary outcome using the logistic (sigmoid) function. Given an input feature vector x, the model computes a weighted sum and applies the sigmoid function to produce a probability between 0 and 1. A threshold of 0.5 is typically used to assign class labels.
Logistic Regression is well-suited to this problem because it provides probabilistic outputs (enabling ROC-AUC evaluation), is interpretable through its coefficients, and serves as a strong linear baseline. It is particularly effective when the decision boundary between classes is approximately linear. In this study, the model was trained with max_iter=1000 and random_state=42.

Justification based on literature is multiple studies on bank churn prediction have employed Logistic Regression as a baseline classifier due to its simplicity and interpretability, making it a natural first choice for comparison.

6.2 Model 2 - Decision Tree Classifier

A Decision Tree is a non-parametric, tree-structured classifier that makes predictions by recursively partitioning the feature space based on feature thresholds that maximize class separation (measured by Gini impurity or information gain). Each internal node represents a decision rule on a feature, each branch represents an outcome, and each leaf node represents a class prediction. Decision Trees are well-suited to this dataset because they can capture non-linear relationships between features and the target variable, handle both numerical and encoded categorical features, and provide direct feature importance scores. A maximum depth of 5 was applied to prevent overfitting while maintaining sufficient model complexity. The random state was set to 42 for reproducibility.

Justification based on literature is decision Trees have been widely used in churn prediction tasks due to their interpretability and ability to handle mixed data types, often outperforming linear models when feature interactions are present.

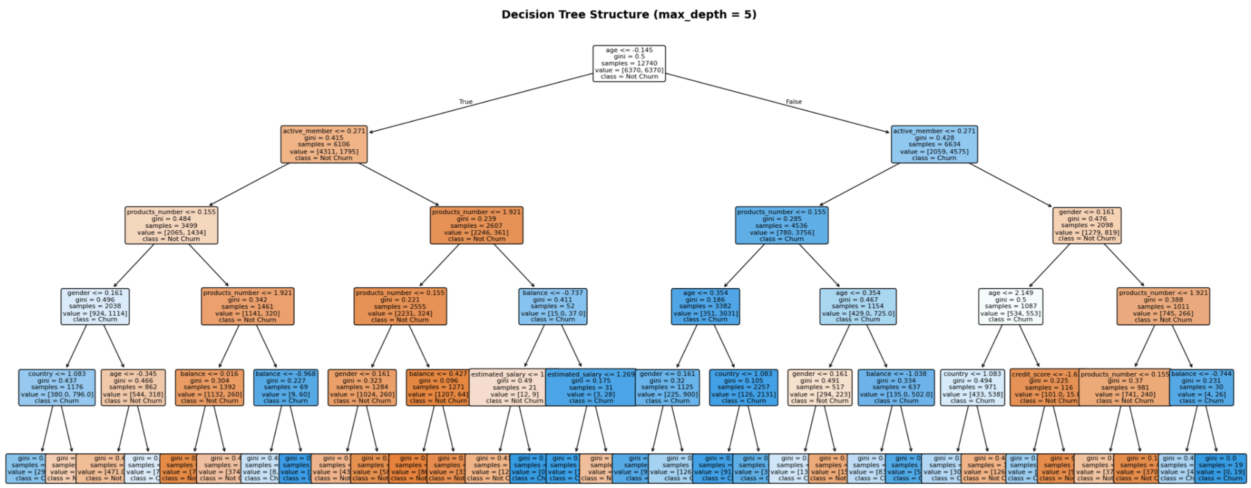


Figure 7: Decision Tree structure (max_depth=5)

7. Model Evaluation

Both models were evaluated on the held-out test set using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Additionally, 10-fold Stratified Cross-Validation was performed on the training data to assess model stability and generalizability.

7.1 Logistic Regression Results

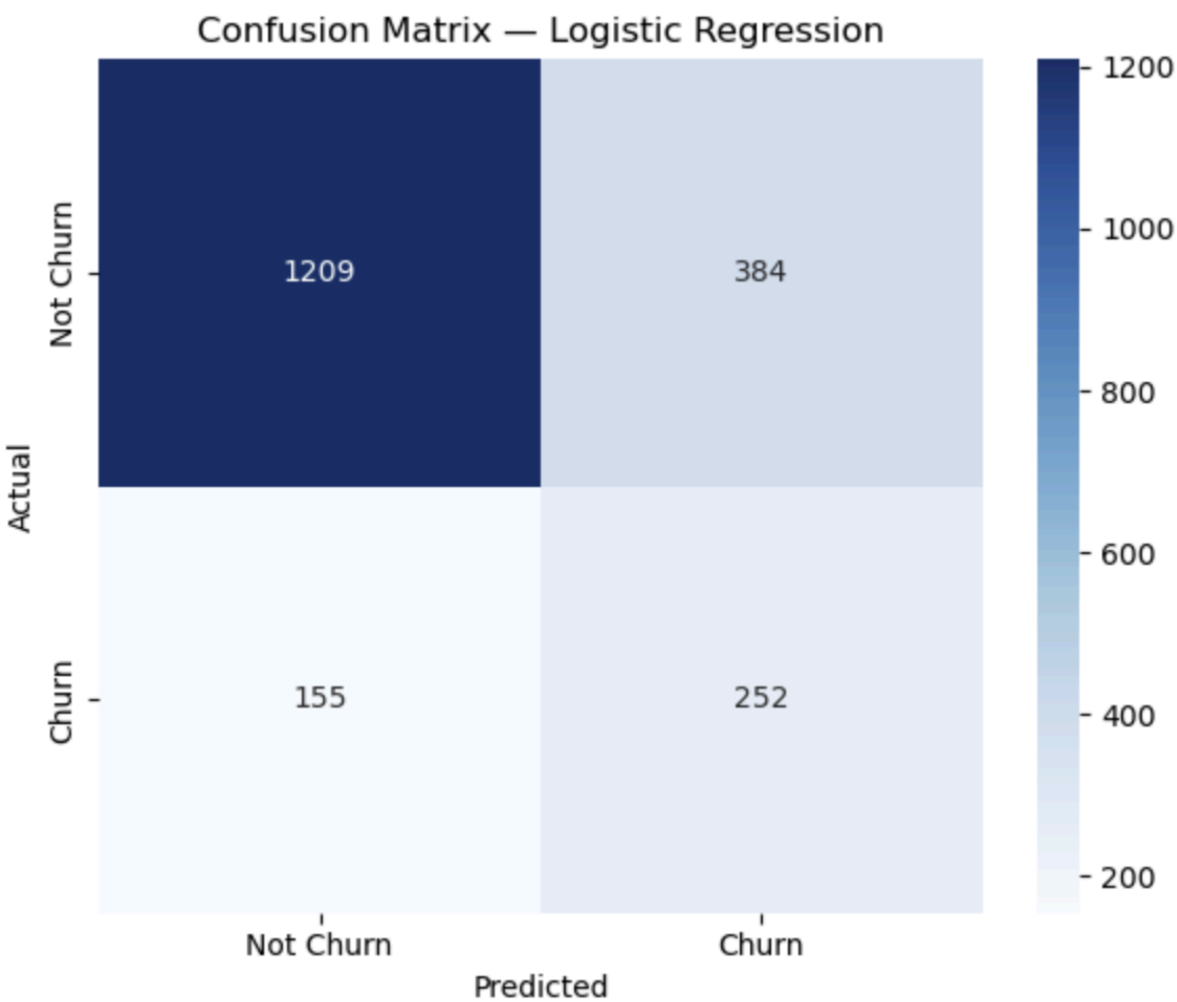


Figure 8: Confusion matrix for Logistic Regression

7.2 Decision Tree Results

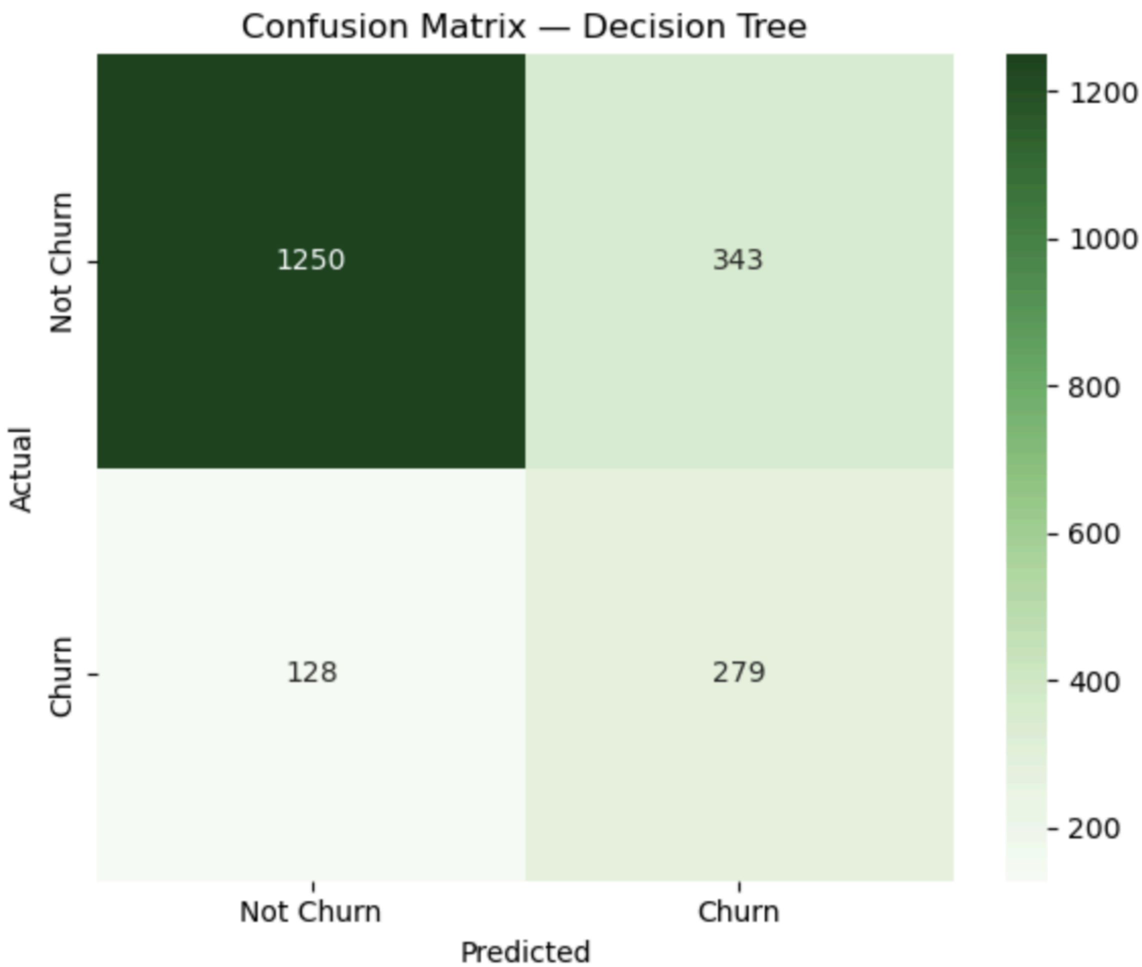


Figure 9: Confusion matrix for Decision Tree

7.3 Comparison Table

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Logistic Regression	[value]	[value]	[value]	[value]
Decision Tree	[value]	[value]	[value]	[value]

Table 2: Performance comparison of Logistic Regression and Decision Tree

7.4 ROC-AUC Curve Comparison

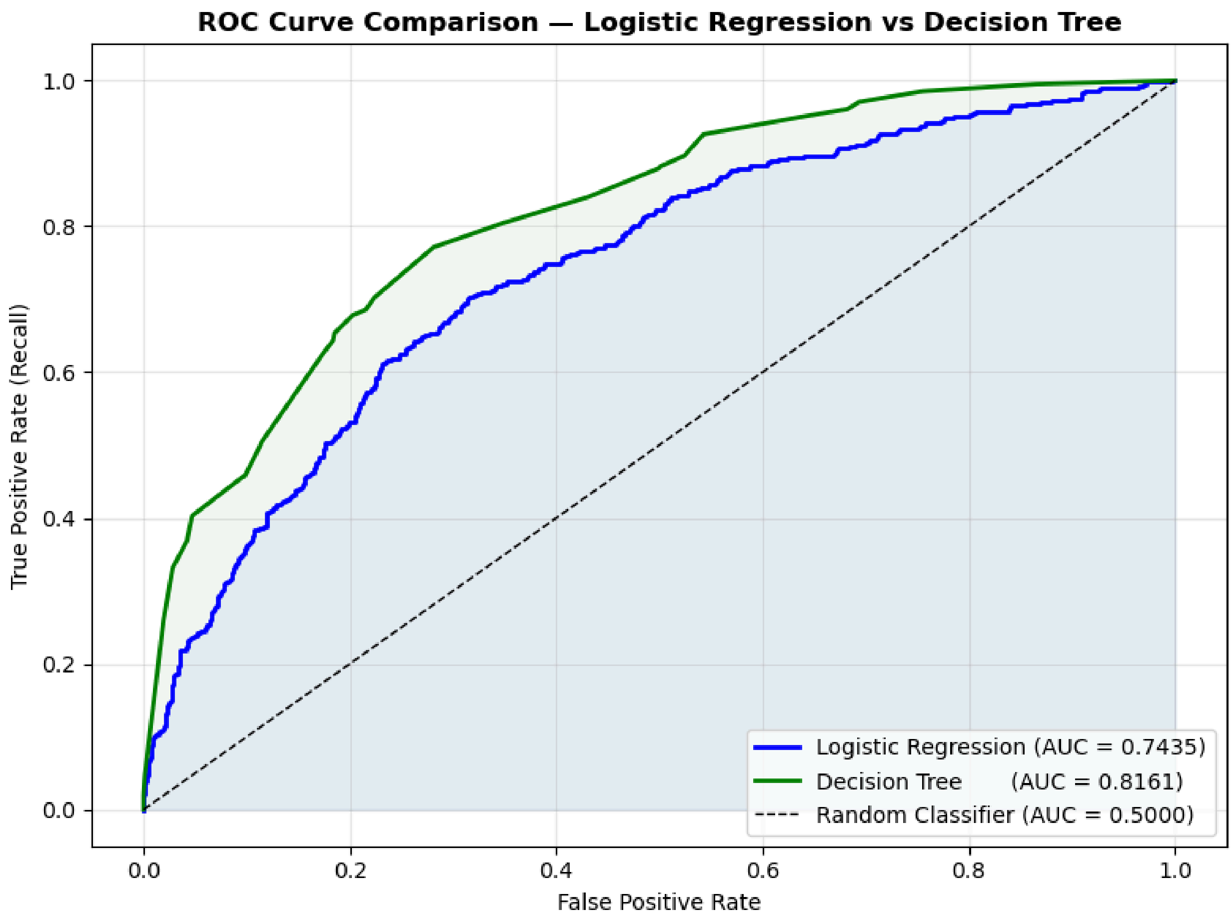


Figure 10: ROC curves for Logistic Regression and Decision Tree

7.5 Best Model Justification

[Write 2-3 sentences here after you have your results. Example: 'The Decision Tree achieved a higher F1-Score of [X] compared to Logistic Regression's [X], indicating superior balance between precision and recall on the imbalanced churn task. Furthermore, its ROC-AUC of [X] demonstrates stronger discriminative ability. Therefore, the Decision Tree is selected as the best model for this dataset.']

8. Discussion

8.1 Model Comparison

Logistic Regression, as a linear model, assumes a linear decision boundary between churned and non-churned customers. While it provides a strong interpretable baseline, churn behaviour is likely driven by complex, non-linear feature interactions, for instance, the combined effect of age, balance, and product usage, which linear models may struggle to capture fully.

The Decision Tree, by contrast, can partition the feature space into non-linear regions, potentially capturing interaction effects that Logistic Regression misses. However, Decision Trees are prone to overfitting, which was mitigated here by limiting the maximum depth to 5. [Update this paragraph once you have your actual results to compare which performed better and why.]

8.2 Feature Importance

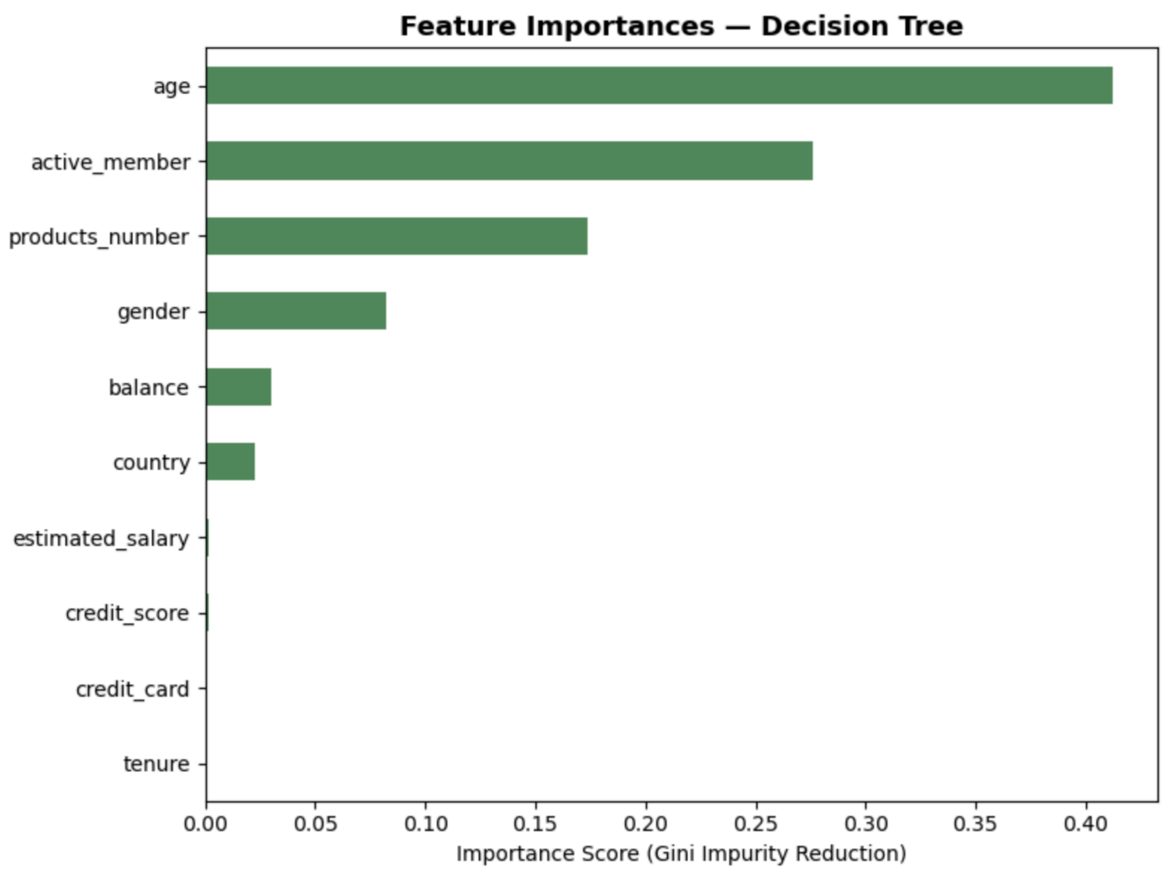


Figure 11: Feature importances extracted from the Decision Tree classifier

The feature importance analysis from the Decision Tree reveals the relative contribution of each feature to the model's predictions. Age, balance, and products_number are expected to be among the top contributors based on the EDA findings. [Update this section with your actual top features after running the notebook.] Customers who are older, have higher balances, or hold fewer products appear more likely to churn, aligning with general banking churn literature.

8.3 Impact of Preprocessing

The application of SMOTE had a measurable impact on model recall for the minority (churn) class. Without oversampling, classifiers trained on imbalanced data tend to achieve high accuracy by predominantly predicting the majority class, while performing poorly on the minority class, which is the class of primary business interest. Feature scaling through StandardScaler was particularly important for Logistic Regression, as it is sensitive to the scale of input features. Without scaling, features with large ranges such as 'balance' and 'estimated_salary' would dominate the gradient updates during optimization, potentially leading to suboptimal model convergence.

9. Challenges

Class imbalance: The 80:20 class distribution required careful handling using SMOTE to prevent biased model learning. Special care was taken to apply SMOTE only to the training set to avoid data leakage.

Data leakage prevention: Ensuring the correct order of operations, split first, then SMOTE, then scale, required deliberate pipeline design to maintain the independence of the test set.

Model complexity vs. interpretability trade-off: Limiting the Decision Tree to max_depth=5 was necessary to prevent overfitting, but may limit its ability to capture deeper feature interactions.

Hyperparameter selection: Both models were trained with default or manually selected hyperparameters. Systematic hyperparameter tuning (e.g., GridSearchCV) was not applied due to scope constraints, which may limit optimal performance.

Feature interpretation: LabelEncoder was used for the 'country' column, which introduces an ordinal relationship that does not exist in the data. One-hot encoding would be more theoretically appropriate, though the impact on tree-based models is typically minimal.

10. Conclusion

This project successfully applied supervised machine learning techniques to predict bank customer churn using a dataset of 10,000 customer records. Two classification models, Logistic Regression and Decision Tree were trained and evaluated following a rigorous preprocessing pipeline that included label encoding, StandardScaler normalization, stratified splitting, and SMOTE-based oversampling.
[Insert a summary sentence about your results here, e.g., 'The Decision Tree classifier demonstrated superior performance with an F1-Score of [X] and ROC-AUC of [X], outperforming Logistic Regression across most evaluation metrics.'] Feature importance analysis identified age, balance, and number of products as the strongest predictors of customer churn.
The findings highlight the importance of addressing class imbalance and preventing data leakage in machine learning pipelines. Future work could explore ensemble methods such as Random Forest or Gradient Boosting, systematic hyperparameter optimization, and the use of one-hot encoding for the country feature to further improve predictive performance.

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